

Python Practice Questions - Extra Set

Topic: While Loops & Lists (Medium Level) | No for loops allowed

■ Program prompt

■ What YOU type

■ Program output

Q1. Movie Ticket Booking System

Level: Medium | Concepts: list, while loop, accumulator, max tracking

A cinema sells tickets at ¥150 each. Write a program to track ticket bookings.

- Keep asking the customer how many tickets they want to book
- Enter 0 to stop taking bookings
- Store each booking amount (tickets x 150) in a list
- Use a while loop to go through the list and find the total revenue
- Also print the highest single booking amount

Program Run (Sample)

```
Enter number of tickets (0 to stop): 2
Enter number of tickets (0 to stop): 5
Enter number of tickets (0 to stop): 1
Enter number of tickets (0 to stop): 0

Total revenue: ¥1200
Highest booking amount: ¥750
```

Q2. Word Frequency Counter

Level: Medium | Concepts: list, while loop, counting/search

Write a program that counts how many times a word appears in a list of entered words.

- Keep asking the user to enter words one at a time
- Store each word in a list
- Type 'done' to stop entering words
- Ask the user for a word to search for
- Use a while loop (not a for loop) to count how many times that word appears

Program Run (Sample)

```
Enter a word (or 'done'): apple
Enter a word (or 'done'): mango
Enter a word (or 'done'): apple
Enter a word (or 'done'): banana
```

Enter a word (or 'done'): **apple**

Enter a word (or 'done'): **done**

Enter word to search: **apple**

'apple' appears 3 times

Q3. Temperature Monitoring System

Level: Medium | Concepts: list, while loop, min/max tracking, conditionals

A weather station records temperature readings. Write a program to analyze them.

- Keep asking the user to enter temperature readings (in C) one by one
- Store each reading in a list
- Type 'stop' to end input
- Use a while loop to find the highest and lowest temperature
- Also count how many readings were above 35C (heat warning)

Program Run (Sample)

Enter temperature in C (or 'stop'): **30**

Enter temperature in C (or 'stop'): **38**

Enter temperature in C (or 'stop'): **22**

Enter temperature in C (or 'stop'): **40**

Enter temperature in C (or 'stop'): **28**

Enter temperature in C (or 'stop'): **stop**

Highest: 40

Lowest: 22

Heat warnings (>35C): 2

Q4. Library Book Return Fine Calculator

Level: Medium | Concepts: list, while loop, accumulator, conditional counting

A library charges ¥5 per day for late returns. Write a program to manage fines.

- Keep asking the user to enter the number of overdue days for a book
- Enter -1 to stop
- For each entry, calculate the fine (0 if days is 0 or negative, but not -1)
- Store each fine amount in a list
- Use a while loop to print the total fines collected and how many books had no fine

Program Run (Sample)

Enter overdue days (-1 to stop): **3**

Enter overdue days (-1 to stop): 0

Enter overdue days (-1 to stop): 7

Enter overdue days (-1 to stop): -1

Total fines collected: ¥50

Books with no fine: 1
